THE NORMALIZATION IN A REDUCED SCHEME IS A FINITE MORPHISM OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof ends with the author's own 'Some small details omitted.' after the two uses of local algebraicity; the sentence is reproduced verbatim. The hypotheses used are named explicitly, so the omission is bounded.

Editorial note. This card was re-admitted after a gate audit: a text-similarity screen had flagged it against a statement of another chapter whose conclusion is that the normalization is reduced rather than finite. The two are different propositions, so the ruling recorded for this lane is to keep this card.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), which since 2026-10-12 also carries the project's own reference/slogan/history definitions.

Allowed standard prerequisites (editorial).

- Locally algebraic schemes over a field are locally Noetherian and quasi-separated. Properties, Lemma properties-lemma-locally-Noetherian-quasi-separated and Morphisms, Lemma morphisms-lemma-finite-type-noetherian, invoked at proof lines 4978-4981. Supplies that Y is quasi-separated, hence that the morphism is quasi-separated.
- Quasi-separatedness is stable under composition with a separated morphism. Schemes, Lemma schemes-lemma-compose-after-separated, invoked at proof lines 4981-4982. Supplies that the morphism f is quasi-separated.
- Definition of the normalization of a scheme in another scheme. Morphisms, Definition morphisms-definition-normalization-X-in-Y, invoked at proof lines 4983-4984. Supplies the object X' whose finiteness is at stake.
- Finiteness of normalization in a reduced scheme over a Nagata base. Morphisms, Lemma morphisms-lemma-nagata-normalization-finite-general and Lemma morphisms-lemma-ubiquity-nagata, invoked at proof lines 4984-4989. Supplies the finiteness conclusion.

Source reference. Original definition 035H: sources/stacks-chow/tags.txt, source lines 4128–4128. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0AVK: sources/stacks-chow/tags.txt, source lines 13688–13688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035B: sources/stacks-chow/tags.txt, source lines 4122–4122. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OY: sources/stacks-chow/tags.txt, source lines 2211–2211. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KV: sources/stacks-chow/tags.txt, source lines 2064–2064. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let $f : Y \to X$ be a quasi-compact morphism of locally algebraic schemes over k. Let X' be the normalization of X in Y. If Y is reduced, then $X' \to X$ is finite.*

Proof. Since Y is quasi-separated (by Properties, Lemma 01OY and Morphisms, Lemma 01T6) the morphism f is quasi-separated (Schemes, Lemma 01KV). Hence Morphisms, Definition 035H applies. The result follows from Morphisms, Lemma 0AVK. This uses that locally algebraic schemes are locally Noetherian (hence have locally finitely many irreducible components) and that locally algebraic schemes are Nagata (Morphisms, Lemma 035B). Some small details omitted. □